

CSC1042 Computer Network II

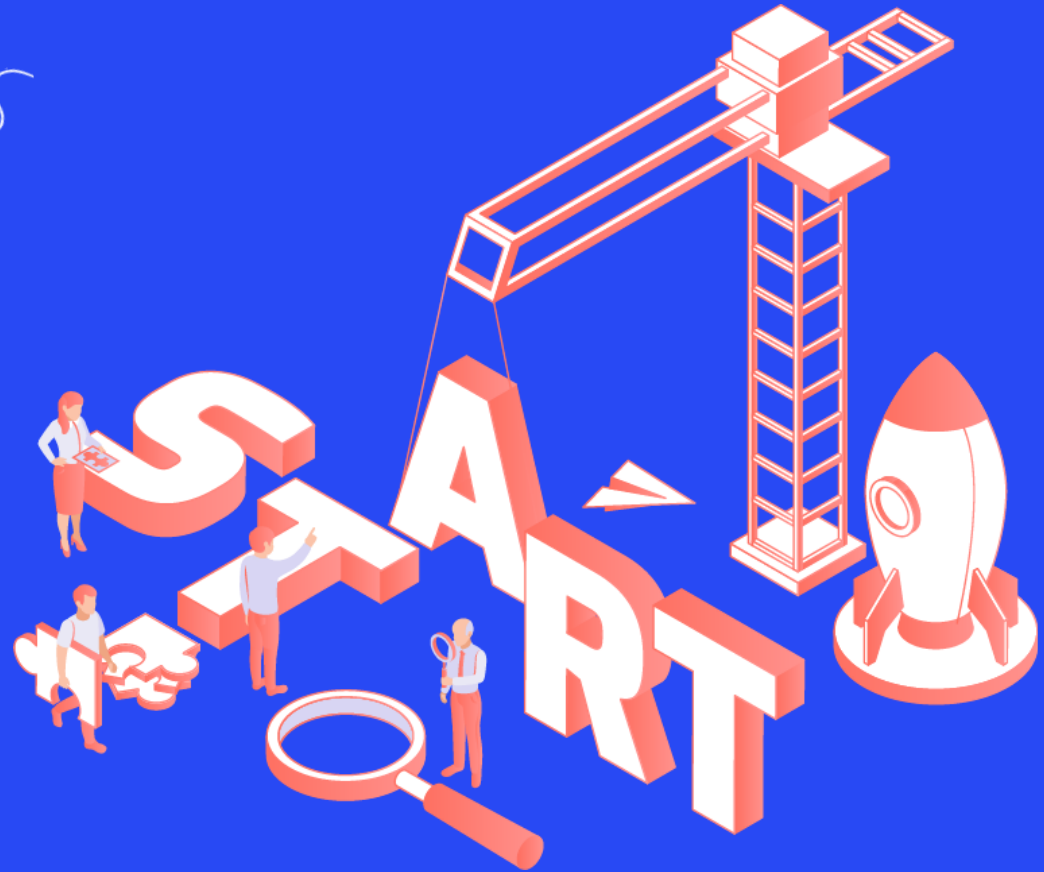
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IPv4
addressing



learning Outcomes

- Describe the IPv4 address structure
- Compare the characteristics and uses of unicast, broadcast, and multicast IP addresses
- Explain public, private, and reserved IP address
- Explain the use of NAT and how it works



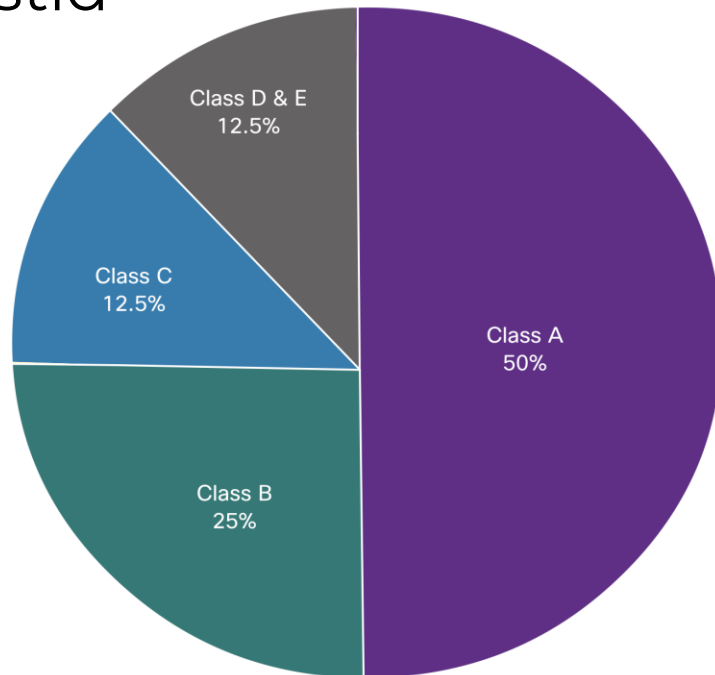
IPv4 Addresses

- 32-bit long addresses
- Each IP address is made up of:
 - *Network portion (Netid): indicating the network to which the IP address belongs*
 - *Netid should be identical for all devices connected to the same network*
 - *Host portion (Hostid): identifies a specific host within the network*
 - Hostid should be unique



IPv4 classes

- 5 classes: A,B,C,D, and E
- The size of **Netid** and **Hostid** varies across the classes
- Classes D and E do not have netid and hostid



Class A

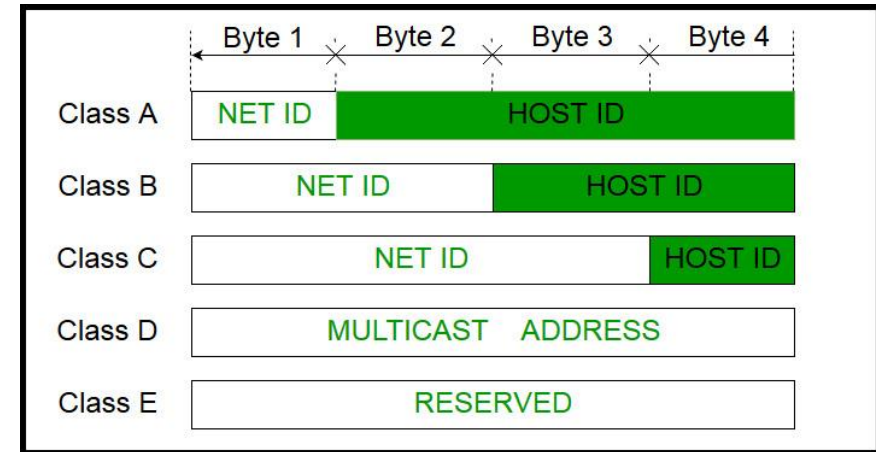
Total Networks: 128
Total Hosts/Net: 16,777,214

Class B

Total Networks: 16,384
Total Hosts/Net: 65,534

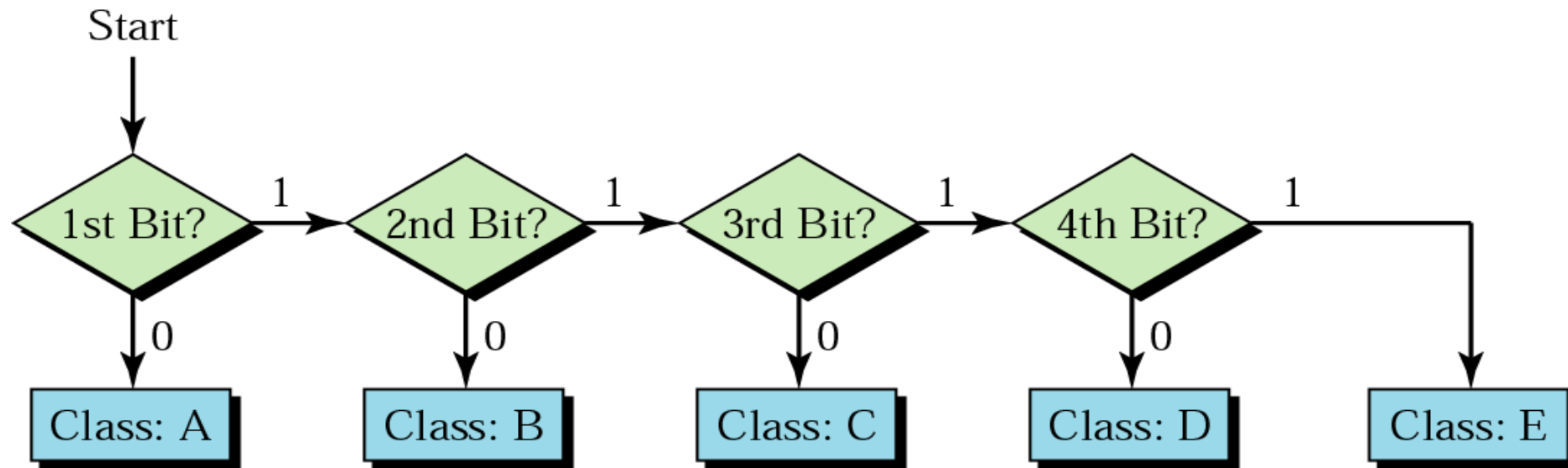
Class C

Total Networks: 2,097,152
Total Hosts/Net: 254



Finding the class of an IPv4 Address

- Convert the left most byte to binary and use the following algorithm
- Start from the left most bit

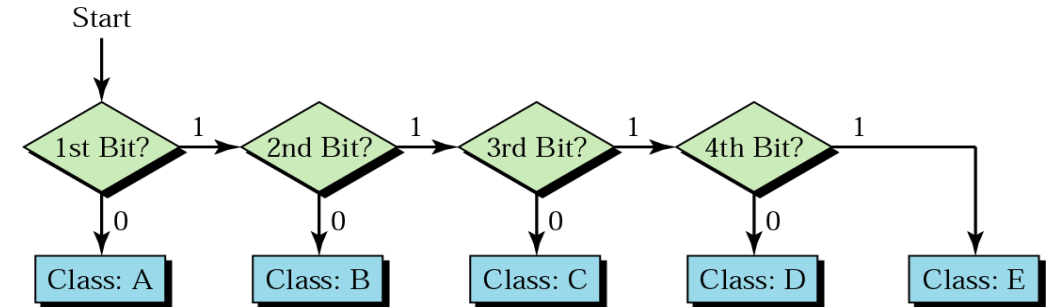


Finding the class of an IPv4 address - Example

- Assume we have the IP address 17.172.224.47
- Convert the first bytes (17 in this case) to binary → 00010001
- Use the algorithm starting from the **left most bit**
- Since the first bit is 0 → this is class A IP address
- Therefore

17.172.224.47

Netid Hostid



Network, Host, and Broadcast addresses

- Within each network, there are 3 types of addresses:
 - **Network address**: the address that identifies the network. It cannot be assigned to hosts
 - **Host addresses**: can be assigned to devices connected to the network
 - **Broadcast address**: used when it is required to reach all devices in a network. It cannot be assigned to hosts

Subnet mask

- Used to identify the network address of an IPv4 address
- It is written as a consecutive sequence of 1s followed by a consecutive sequence of 0s
- The sequence of 1s represents the Netid while the sequence of 0s represents the Hostid
- Note that 11111111 in binary is 255 in decimal

	<u>Network</u>	<u>Host</u>	<u>Host</u>	<u>Host</u>
Class A (1-126) Default subnet mask	255	0	0	0
	<u>Network</u>	<u>Network</u>	<u>Host</u>	<u>Host</u>
Class B (128-191) Default subnet mask	255	255	0	0
	<u>Network</u>	<u>Network</u>	<u>Network</u>	<u>Host</u>
Class C (192-223) Default subnet mask	255	255	255	0

CIDR Notation

- Expressing variable length subnet masks in the dotted decimal representation can become cumbersome
- An alternative method to express a subnet mask is called CIDR
- It specifies the number of bits in NetID

e.g., the IP address 192.168.10.10 with subnet mask 255.255.255.0 can be written as 192.168.10.10/24

Class	Default subnet mask	CIDR notation
A	255.0.0.0	/8
B	255.255.0.0	/16
C	255.255.255.0	/24

Finding the network address

- The actual process of identifying the network address using subnet mask is called **ANDing**

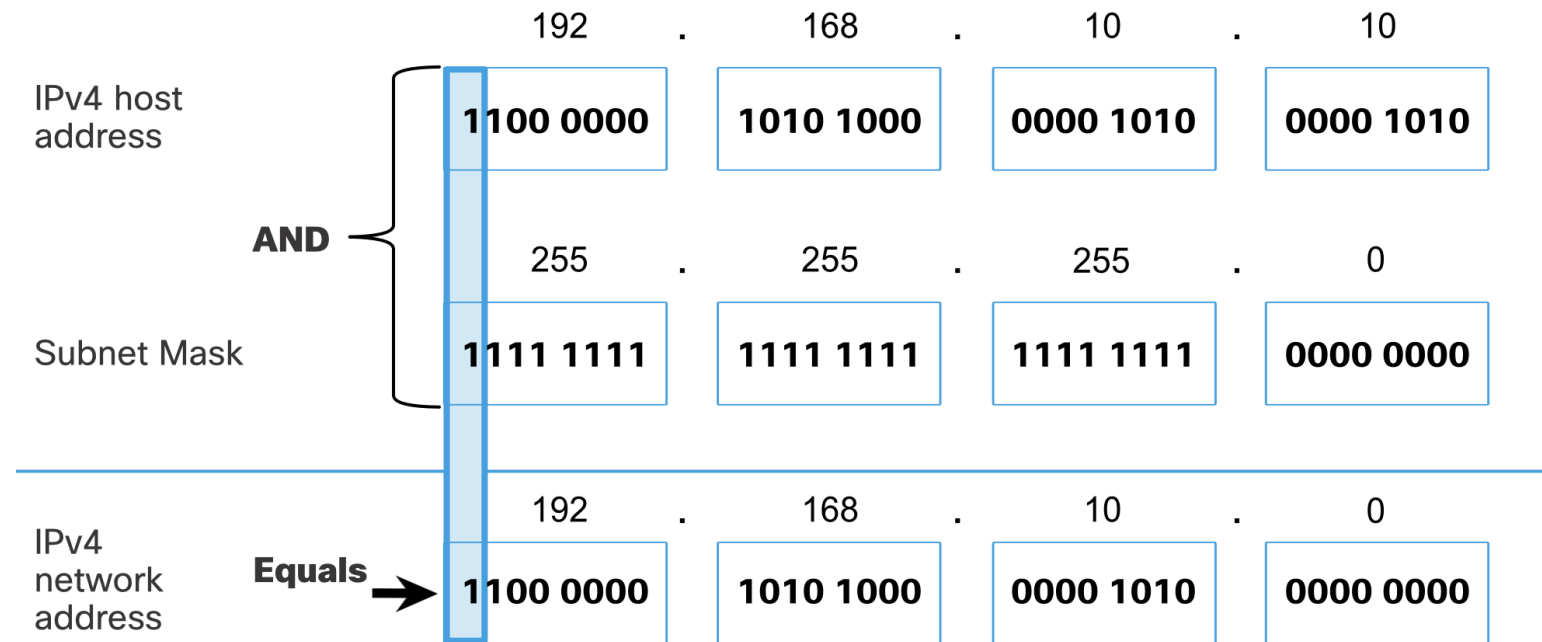
- Refers to the use of the logical AND operator
- Logical AND is the comparison of two bits that produce the results below

- $1 \text{ AND } 1 = 1$
- $0 \text{ AND } 1 = 0$
- $1 \text{ AND } 0 = 0$
- $0 \text{ AND } 0 = 0$

1	0	1	0	1
1	1	1	0	0
1	0	1	0	0

Finding the network address - Example

- Consider a host with IPv4 address 192.168.10.10 and subnet mask 255.255.255.0
- To find the network address, we use the AND operator between the given IP address and the subnet mask



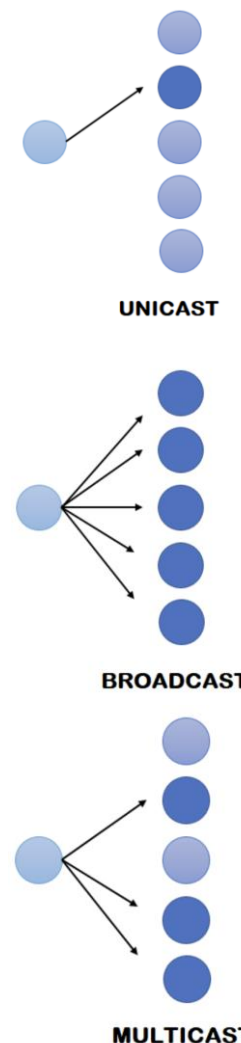
Finding the broadcast address - Example

- Once the network address is identified, set all the bits of the **Hostid** to 1 and covert to decimal
- This will give you the broadcast address of the network

Network address	192	.	168	.	10	.	0
	11000000		10101000		00001010		00000000
Broadcast Address	11000000		10101000		00001010		11111111
	192	.	168	.	10	.	255

Unicast, Broadcast, and Multicast

- There are different ways to send a packet from source to destination, each of which affects the destination IP address
 - **Unicast**: refers to one device sending a packet to one other device. It uses the IP address of that specific device (i.e., host address)
 - **Broadcast**: refers to one device sending a packets to all devices in the network. It uses the broadcast address of the network
 - **Multicast**: refers to one device sending a packet to a set of device in the network. It uses a multicast address (i.e., address from Class D)



How many Host addresses per network?

- To identify the host addresses that can be assigned to devices connected to a network, use the following formula:

$$\text{number of host addresses} = 2^{\text{Hostid bits}} - 2$$

- We subtract 2 addresses that cannot be assigned to any host, which are:
 - The network address
 - The broadcast address

Cheat sheet

Class	Number of Netid bits	Number of Hostid bits	Number of Networks	Number of Hosts per Network	Range
A	8	24	$2^7 = 128$	$2^{24} - 2 = 16,777,214$	0.0.0.0 – 127.255.255.255
B	16	16	$2^{14} = 16,384$	$2^{16} - 2 = 65,534$	128.0.0.0 – 191.255.255.255
C	24	8	$2^{21} = 2,097,152$	$2^8 - 2 = 254$	192.0.0.0-223.255.255.255
D	n/a	n/a	n/a	n/a	224.0.0.0 -239.255.255.255
E	n/a	n/a	n/a	n/a	240.0.0.0- 255.255.255.255

Special IP addresses

- Not all IP addresses are equal
- Some can be used to connect to the Internet – called **public IP addresses**
- Others are used to connect to the local network only – called **private IP addresses** (see table)

IANA Reserved Private Network Ranges	Start of Range	End of Range
24-bit Block	10.0.0.0	10.255.255.255
20-bit Block	172.16.0.0	172.31.255.255
16-bit Block	192.168.0.0	192.168.255.255

- 127.0.0.0 is another special IP address used by workstations for diagnosis purposes
- It's called the **loopback address** (packets sent to this address never reach the network)

THANK YOU